

User-Centered Design for Nonsuicidal Self-Injury Information on TikTok: A Study of Users' Motivations and Circumnavigation of Algorithmic Barriers

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ABSTRACT

Social media platforms can foster community and facilitate harm reduction for individuals engaging in nonsuicidal self-injury or hold these individuals at digital margins. Such platforms include TikTok, a short-form video-sharing social media platform that explicitly bans content depicting, promoting, normalizing, or glorifying activities that could lead to self-harm. As such, TikTok may moderate user-generated nonsuicidal self-injury content, leading to marginalization in this digital space. Using a sequential mixed methods approach comprised of a survey, semi-structured interviews, and participatory design activities, this doctoral research aims to understand users' motivations for adopting TikTok for nonsuicidal self-injury information creation and how TikTok's platform design and content moderation practices can shape users' information creation practices. Findings from this doctoral research will validate the information practices of online nonsuicidal self-injury communities and contribute recommendations on how social media product design teams can better support the motivations of these communities through user-centered design and content moderation policy changes.

CCS CONCEPTS

• Human-center computing; • Collaborative and social computing; Empirical studies in HCI;

KEYWORDS

TikTok, Nonsuicidal self-injury, Marginalization, Content Moderation

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1 INTRODUCTION

Nonsuicidal self-injury (NSSI) refers to the intentional destruction of one's own body tissue without suicidal intent and for purposes

not socially sanctioned [16] and commonly includes behaviors such as cutting, burning, or scratching. Social media platforms afford individuals who engage in NSSI a refuge to build community [14], exchange social support, experience validation with little fear of stigmatization [14], and facilitate harm reduction. Here, harm reduction does not condone NSSI but rather affirms and acknowledges that immediate or permanent abstinence may not be possible or likely for certain individuals. Thus, harm reduction provides individuals autonomy and choice in their recovery by providing practical strategies that maximize safety [22].

Online NSSI communities take the form of communities of practice (i.e., communities based on shared actions or interests rather than places and attributes) [12], enabling users to easily curate the content with which they wish to engage. Over the past decade, research on online NSSI communities has shifted from investigating forums, message boards, and blogs [7] to predominantly focusing on understanding the content on and utilization of popular social media platforms like Instagram [9, 21], Reddit [17], Tumblr [12, 24], Twitter [17], and YouTube [18]. However, the framing of existing research is explicitly divided between the potential benefits and harms of NSSI content [18]. Across these platforms, researchers exploring the benefits of online NSSI communities have suggested that information created and shared within these communities commonly focuses on disclosing experiences of NSSI [12], sharing stories of recovery [24], and seeking and providing emotional support [17]. Alternatively, researchers have positioned the potential harms of online NSSI communities through digital self-harm, which conceptualizes the online communication and activity that leads to, supports, or exacerbates nonsuicidal but intentional harm [20]. However, binarizing NSSI information into either helpful or harmful based on perceived outcomes can venture into deficit framing, as NSSI information is highly contextual. The information created and shared within online NSSI communities reflects the complex and multifaceted needs of individuals engaging in a complex and multifaceted behavior [22]. Thus, binarizing the information created and shared within online NSSI communities overlooks the complexity of NSSI and can work to stigmatize the behavior further.

As existing studies primarily focus on the content within online NSSI communities rather than the context and interdependent nature of the platform design decisions, I aim to address this gap in my doctoral research by examining the impact of platform affordances and constraints in facilitating or hindering content creation of NSSI information on social media platforms, namely TikTok, to which research on NSSI communities has not yet been extended. TikTok is a short-form, video-sharing social media platform designed for users to interact with and through user-generated content. The sociotechnical nature of TikTok exerts power over individuals with

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mental illness, perpetuating adherence to particular norms and values [6], as TikTok specifies that the company moderates content that shows, promotes, or shares plans for self-harm [27]. In this sense, content moderation works to reify notions that specific versions of mental illness are “normal and valued” while rejecting others. As content moderation aims to restrict and influence how people can participate online, users in online NSSI communities must conform to a particular range of norms and values to avoid marginalization on TikTok.

As such, this research focuses on understanding how TikTok’s platform design and content moderation practices can contribute to marginalization and shape the information practices, namely information creation and use, of online NSSI communities. Specifically, this research will jointly consider users’ motivations for adopting TikTok for NSSI information creation and use and the sociotechnical factors that impact these information practices. Through examining TikTok users’ perceptions of the affordances of TikTok’s platform and how those affordances shape information practices, this research can inform product designs that better balance the motivations of NSSI communities and contribute to broader conversations of how social media platforms can better serve marginalized communities and promote inclusivity in digital spaces. This research will address the following research questions:

RQ 1: What are the motivations for TikTok users to create and use nonsuicidal self-injury information?

RQ 2: What barriers do TikTok users encounter when creating and using nonsuicidal self-injury information?

RQ 2.1: How do barriers impact TikTok users’ ability to create and use nonsuicidal self-injury information?

RQ 2.2: How do TikTok users circumnavigate barriers to create and use nonsuicidal self-injury information?

RQ 3: How, if at all, would users redesign TikTok’s algorithm to best support their creation and use information practices?

Through examining users’ motivations for creating and using NSSI information on TikTok, as well as users’ perceived barriers to and circumnavigation of these barriers, these research questions address the current gaps in NSSI literature and will work to provide a better understanding of the impact of TikTok’s platform design on facilitating or hindering NSSI information creation and use.

2 WORK IN PROGRESS

This research will use a sequential mixed methods approach (Figure 1) to explain users’ motivations for creating and using NSSI information on TikTok and users’ experiences with and circumnavigation of algorithmic barriers on the platform. Participants will consist of U.S.-based young adult (18-29 years) TikTok users, as current estimates suggest that between 17% and 38% of adolescents and young adults engage in NSSI [4], and the majority of TikTok users are under the age of 29 [1]. Eligibility criteria will specify that participants have a history of engaging in NSSI (either previously or presently) and are active content creators on TikTok. Through this research, I will address ethical issues through both procedural ethics and ethics in practice [26] using guidelines established by the Association of Internet Researchers [8]. In doing so, I will consider ethics of care [8], which reflects on the nature of knowledge, including situated knowledge, which asks what counts as knowledge and

how this knowledge is gained [13]. Thus, ethics of care positions that knowledge in itself is neither good nor bad but is good or bad in relation between its content and context.

This research will comprise of two phases, both of which have been informed by two pilot studies. In the first phase, approximately 100-150 users within NSSI communities on TikTok will complete an online survey addressing users’ motivations for creating and using NSSI information on TikTok and the barriers they encounter on the platform. A combination of purposeful sampling strategies, including stratified and snowball sampling to develop networked relations with community contacts, will be used to recruit TikTok users within online NSSI communities. I will identify community contacts using accounts with a high following base I collected during a previously conducted content analysis of 150 NSSI TikTok videos [19]. Community contacts can provide additional points of contact, disseminate information about the study through their personal trusted channels, and volunteer to participate. At the conclusion of the survey, participants will be asked if they would like to participate in a follow-up interview. Descriptive statistics will be used to summarize nominal variables, and chi-square tests will be used to measure ordinal responses. Additionally, the open-ended survey responses will be analyzed through inductive coding [3]. Conceptual themes elicited from open coding will be used to supplement and organize statistical findings and inform the interview protocol for the second phase.

The second phase of the study will include virtual semi-structured interviews with a participatory design activity. I will select approximately 15-20 survey respondents to participate in a synchronous or asynchronous semi-structured interview, as has proven effective in previous studies with individuals who engage in NSSI [23]. The interview protocol will be informed by both the survey results and a pilot study that was previously conducted, wherein I facilitated eight semi-structured interviews with users within NSSI communities on TikTok. The interview protocol will expand on participants’ experiences with creating NSSI content on TikTok, including users’ motivations for creating NSSI content, perceptions of content moderation policies, and strategies for circumnavigating content moderation to maintain visibility on TikTok. In the latter half of the interview, participants will engage in a participatory design activity focused on future-making. Future-making activities include brainstorming a list of critiques of a current situation (i.e., challenges NSSI communities face on TikTok) and then transforming those critiques into positive outcomes [2]. The participatory design activity, which aims to capture how users would redesign TikTok to best support their information creation and use practices, will consist of three steps: identifying problems, envisioning design ideas, and concretizing ideas through sketching.

Data elicited from interviews will be analyzed using the constant comparison method [10], where data analysis will be informed by the results of survey responses and the results of the pilot studies (i.e., content analysis and pilot interviews). Following preliminary coding, I will use focused and axial coding to summarize and condense the codebook before re-coding the data, allowing once more for the identification of emergent coding patterns and categories to occur [25]. Throughout the coding process, I will write analytic memos [25] to document and reflect on code choices, clustering, and emergent patterns, themes, and concepts. After the preliminary

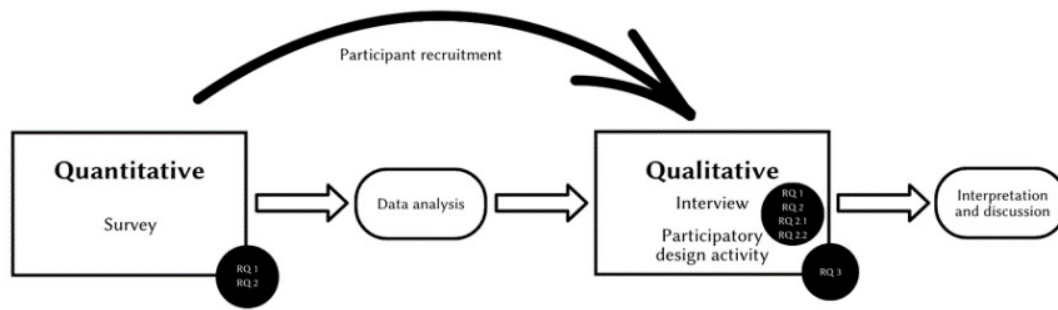


Figure 1: Sequential mixed methods research design

analysis, participants will receive a summary of the findings and will be invited to member-check their interview to test the integrity of findings and establish credibility.

3 EXPECTED CONTRIBUTIONS

Findings from this research will have implications for both the use and design of social media platforms. Specifically, expected contributions will center on validating the information practices of NSSI communities and will focus on how social media product design teams can better support the motivations of marginalized communities through user-centered design. Further, this doctoral research will contribute recommendations for content moderation policy change.

First, this doctoral research will contribute design recommendations that appreciate the individual impact of technological “solutions,” which conflict with current broad, population-level approaches to mental illness to be categorized as harmful content [6], which represents a restriction of expression. Design affordances, such as content moderation, perpetuate social control structures by focusing on identifying and mitigating information practices associated with NSSI. However, incorporating the subjective experiences of NSSI is an important component of understanding and validating the information produced by users in NSSI communities. By contextualizing the information practices of NSSI communities, product design teams can better understand the “normalcy and disruption for the individual” [6] and attend to the ways that design for these communities fails. In this sense, product design teams should not obsessively focus on “fixing” NSSI, such as solely providing mental health resources or erasing its existence entirely, but instead should focus on supporting users’ experiences. Product design teams may accomplish this by integrating community members’ voices into the design process. In doing so, product designers can employ a stress case approach to highlight the sociocultural discourses embedded into technologies and account for the marginalizing outcomes produced by technological affordances [15].

Finally, this doctoral research will contribute recommendations for content moderation policy change. As it stands, TikTok may be failing users within NSSI communities via their Community Guidelines. TikTok presents issues of autonomy within the bounded sphere of their Community Guidelines, wherein users’ autonomy

and consent are undermined [11]. In this sense, TikTok fails to acknowledge the duty of care to enable destigmatized conversations of NSSI within context and allow users to express and share experiential knowledge within communities of practice. As such, this research will include recommendations that TikTok’s content moderation policies should consider NSSI content within context to ensure the platform can function as a safe, destigmatized space to engage in discussions of NSSI.

4 CONCLUSION

The study will detail how TikTok users work to resist content moderation to establish a safe and destigmatized space for themselves and other users who are engaging in or recovering from NSSI. While TikTok states that they encourage supportive interactions among people with mental illnesses, the sociotechnical nature of the platform can lead to moderation and marginalization of online NSSI communities. Thus, this research aims to inform a redesign of sociotechnical systems to more effectively and transparently balance the needs of online NSSI communities on social media platforms.

REFERENCES

- [1] Brooke Auxier and Monica Anderson. 2021. Social media use. Retrieved April 15, 2023 from <https://www.pewresearch.org/internet/2021/04/07/social-media-use-in-2021/>
- [2] Eva Brandt, Thomas Binder, and Elizabeth B-N. Sanders. 2012. Tools and techniques: Ways to engage telling, making and enacting. In Jesper Simonsen and Toni Robertson (Eds.) *Routledge International Handbook of Participatory Design* (1 ed.), 145-181. Routledge.
- [3] Kathy Charmaz. 2014. *Constructing grounded theory*. Sage Publishing, Inc.
- [4] Annarosa Cipriano, Stefania Cella, and Paolo Cotrufo. 2017. Nonsuicidal self-injury: A systematic review. *Frontiers in Psychology* 8, 1-14. <https://doi.org/10.3389/fpsyg.2017.01946>
- [5] Jessica L. Feuston and Anne Marie Piper. 2019. Everyday experiences: small stories and mental illness on Instagram. In *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems*, 1-14. <https://doi.org/10.1145/3290605.3300495>
- [6] Jessica L. Feuston, Alex S. Taylor, and Anne Marie Piper. 2020. Conformity of eating disorders through content moderation. In *Proceedings of the ACM on Human-Computer Interaction* 4, CSCW1, 1-28. <https://doi.org/10.1145/3392845>
- [7] Anna Gradin Franzén, and Lucas Gottzén. The beauty of blood? Self-injury and ambivalence in an Internet community. 2011. *Journal of Youth Studies* 14, 3, 279-294.
- [8] aline shakti franzke, Anja Bechmann, Michael Zimmer, Charles M. Ess. 2020. Internet research: Ethical guidelines 3.0. Retrieved April 15, 2023 from <https://aoir.org/reports/ethics3.pdf>
- [9] Jory A. Fulcher, Sarah Dunbar, Elizabeth Orlando, Sarah J. Woodruff, and Sara Santarossa. 2020. #Selfharm on Instagram: Understanding online communities surrounding non-suicidal self-injury through conversations and common properties

- among authors. *Digital Health* 6, 1-13. <https://doi.org/10.1177/2055207620922389>
- [10] Barney G Glaser, and Anselm L. Strauss. 2017. *The discovery of grounded theory: Strategies for qualitative research*. Routledge.
- [11] Joanne Gray and Alice Witt. 2021. A feminist data ethics of care framework for machine learning: The what, why, who and how. *First Monday* 26, 12. <https://doi.org/10.5210/fm.v26i12.11833>
- [12] Federica Guccini, and Gerald McKinley. 2022. "How deep do I have to cut?": Non-suicidal self-injury and imagined communities of practice on Tumblr. *Social Science & Medicine* 296, 1-9. <https://doi.org/10.1016/j.socscimed.2022.114760>
- [13] Donna Haraway. 1988. Situated knowledges: The science question in feminism and the privilege of partial perspective. *Feminist Studies* 14, 3, 575-599.
- [14] Ray Jones, Siobhan Sharkey, Tamsin Ford, Tobit Emmens, Elaine Hewis, Janet Smithson, Bryony Sheaves, and Christabel Owens. 2011. Online discussion forums for young people who self-harm: User views. *Psychiatrist* 35, 10, 364–368. <https://doi.org/10.1192/pb.bp.110.03344>
- [15] Vanessa L. Kitzie. 2019. "That looks like me or something I can do": Affordances and constraints in the online identity work of US LGBTQ+ millennials. *Journal of the Association for Information Science and Technology* 70, 12, 1340-1351. <https://doi.org/10.1002/ASI.24217>
- [16] David E. Klonsky, Sarah E. Victor, and Boaz Y. Saffer. 2014. Nonsuicidal self-injury: What we know, and what we need to know. *The Canadian Journal of Psychiatry* 59, 11, 565-568. <https://doi.org/10.1177/07067437140590110>
- [17] Anna Lavis and Rachel Winter. 2020. #Online harms or benefits? An ethnographic analysis of the positives and negatives of peer-support around self-harm on social media. *Journal of Child Psychology and Psychiatry* 61, 8, 842-854. <https://doi.org/10.1111/jcpp.13245>
- [18] Stephen P. Lewis and Amanda K. I. Knoll. 2015. Do it yourself: Examination of self-injury first aid tips on YouTube. *Cyberpsychology, Behavior, and Social Networking* 18, 5, 301-304. <https://doi.org/10.1089/cyber.2014.0407>
- [19] Valerie Lookingbill. 2022. Examining nonsuicidal self-injury content creation on TikTok through qualitative content analysis. *Library & Information Science Research* 44, 4, 1-8. <https://doi.org/10.1016/j.lisr.2022.101199>
- [20] Jessica Pater and Elizabeth Mynatt. 2017. Defining digital self-harm. In *Proceedings of the 2017 ACM Conference on Computer Supported Cooperative Work and Social Computing*. 1501-1513.
- [21] Jacobo Picardo, Sarah K. McKenzie, Sunny Collings, and Gabrielle Jenkin. 2020. Suicide and self-harm content on Instagram: A systematic scoping review. *PLoS One* 15, 9, e0238603. <https://doi.org/10.1371/journal.pone.0238603>
- [22] Emma G Preston and Amy E. West. 2022. Straight to the source: E-communities for nonsuicidal self-injury and the emerging case for harm reduction in the treatment of nonsuicidal self-injury. *Clinical Psychological Science* 10, 4, 801-813.
- [23] Yukari Seko, Sean A. Kidd, David Wiljer, and Kwame J. McKenzie. 2015. On the creative edge: Exploring motivations for creating non-suicidal self-injury content online. *Qualitative Health Research*, 25, 10, 1334-1346. <https://doi.org/10.1177/104973231557013>
- [24] Yukari Seko and Stephen P. Lewis. 2018. The self – harmed, visualized, and reblogged: Remaking of self-injury narratives on Tumblr. *New Media & Society* 20, 1, 180-198. <https://doi.org/10.1177/1461444816660783>
- [25] Johnny Saldaña. 2021. *The coding manual for qualitative researchers* (4th ed.). SAGE.
- [26] Siobhan Sharkey, Ray Jones, Janet Smithson, Elaine Hewis, Tobit Emmens, Tamsin Ford, and Christabel Owens. 2011. Ethical practice in internet research involving vulnerable people: lessons from a self-harm discussion forum study (SharpTalk). *Journal of Medical Ethics* 37, 12, 752-758.
- [27] TikTok. 2023. Community Guidelines Enforcement Report. (March 2023). Retrieved July 6, 2023 from <https://www.tiktok.com/transparency/en-us/community-guidelines-enforcement-2022-2/>